

K55: Heegner Walk-Back Audit — Criteria for Identifying L1-Source Claims That Need Tier-Review

Keeper (filing per Cal scoping, Option C governance)

2026-05-19 Tuesday morning

K55: Heegner Walk-Back Audit

Context

Sunday May 17 saw the Heegner-Stark Root #7 promotion (K47) via Deuring 1941 CM theory + Cremona 49a1 mechanism. Subsequently a broader proposal emerged: “9 Heegner discriminants as L1 source” en masse — promoting Heegner discriminants themselves as a classical-integer-set L1 source, separate from K47.

Grace self-withdrew the broader proposal Sunday morning (“Heegner discriminants don’t yet match the source-theorem signature in the way Klein/VSC/K3 Hodge/Wallach do”). The team’s vote on Sunday initially restored the broader claim too quickly; Cal walked it back partially on Sunday afternoon, aligning with Grace’s structural reasoning.

The walk-back raised a meta-question: what other classical-integer-set claims have we been treating as L1 source when they’re really I-tier empirical patterns? Cal scoped this question Tuesday morning at Casey’s directive (Option C: Cal scope + Keeper K-audit + team review).

This K55 audit ratifies Cal’s four-criteria framework and assigns the tier-A/B/C lists to follow-up actions.

Cal’s four walk-back criteria (W1-W4)

A classical-integer-set claim warrants walk-back review if ALL four fire:

- W1: Classical integer set identified (e.g., “9 Heegner discriminants {1, 2, 3, 7, 11, 19, 43, 67, 163}”).
- W2: BST-decomposability is the primary evidence (e.g., “4 of 9 are BST primaries; the others factor via BST products in specific observables”).

- W3: Mechanism chain to D_{IV}^5 is missing OR is BST-internal (no published-classical-math mechanism exhibited; only “BST primaries appear in this set” as observation).
- W4: Tier label is L1-source-shaped (e.g., “Root #N”, “established L1 classical source”) despite W3.

If all four fire $\rightarrow$ walk-back review. If any one doesn't fire $\rightarrow$ no walk-back needed.

I ratify W1-W4 as the standing criteria for identifying classical-integer-set L1-source claims needing tier-review.

Tier-A clean list (9 items) — RATIFIED, no walk-back needed

Cal identified 9 current ESTABLISHED roots that pass criterion test cleanly (they have published-classical-mathematics mechanism chains exhibited, so W3 does NOT fire):

K-audit	Root	Classical mechanism
K43	VSC	Von Staudt-Clausen 1840 — Bernoulli denominator structure
K3 Hodge	K3 Hodge	K3 1962/64 — Hermitian symmetric domain Hodge classes
Wallach	Wallach 1976	Wallach K-types on rank-2 Hermitian symmetric spaces
K47	Klein A_5	Klein 1884 — $A_5 \subset SO(5)$ $\subset K(D_{IV}^5)$ embedding
K45	Mathieu	Mukai 1988 + EOT 2010 — $M_{23} \subset$ $Aut_{symp}(K3)$ embedding
K47	Heegner-Stark via 49a1	Deuring 1941 CM theory anchored at Cremona 49a1 (DISTINCT from “9 discriminants en masse”)
K48	Conway	Duncan 2007 — $Aut(V^{\{f\}}) = Co_0$ at $c=12$
K46	Goeppert Mayer	Mayer-Jensen 1949 — $SU(2)_J \times SO(3) \subset SO(5)$
Borcherds/McKay	Mechanisms	Borcherds 1992 / McKay 1979 published constructions

All 9 have classical-mechanism chains exhibited. W3 does NOT fire. No walk-back needed.

The K45-K48 audit chain already supplied what Grace’s Sunday walk-back identified as missing for K47-broader. The discipline worked: Architecture Day correctly distinguished K47-Heegner-Stark (mechanism: Deuring CM via 49a1) from the would-be “9 discriminants en masse” claim (no mechanism).

Tier-B verify list (4 items) — SPOT-CHECK RECOMMENDED

Cal flagged 4 items for spot-check verification. These probably pass but warrant confirmation:

Item	Mechanism status	Verification scope
Ogg via Borchers	Mechanism appears at Borchers 1992	Confirm specific theorem chain; ~30 min
Goeppert Mayer (K46)	Mayer-Jensen 1949	Confirm $SU(2)_J \times SO(3)$ embedding chain holds; ~30 min
Mathieu via EOT	EOT 2010 + Mukai 1988	Confirm M_{23} embedding is genuinely L1-source structure not L1.5-mechanism; ~30 min
Bravais L1-mediated	Bravais 1849 / Frankenheim 1842 + crystallographic restriction	K50 Option C ruling — confirm “L1-mediated” tier holds (not full L1, not I-tier observation); ~30 min

Action: Grace cataloging support to flag any tier-B verifications that need attention. Cal triangulation on any specific spot-check Keeper or Grace requests. Estimated 1-2 hours total across the four items.

Tier-C walk-back list (3 candidates) — REFERRED TO K56, K57, K58 (or similar)

Cal identified 3 candidates where walk-back review is warranted. Each becomes its own K-audit:

K56 candidate: Cathedral component-level Δr table (Cal flag a)

Section 7.1 / Toy 3012 catalogs the Cathedral component-level Δr decomposition with several 1-2% identifications labeled D-tier. Cal flag: these identifications lack

exhibited mechanism chains to D_IV⁵. Walk-back recommendation: downgrade to I-tier unless chains are supplied per-item.

Audit scope: enumerate each Δr identification (probably 6-8 entries), check whether mechanism chain is filed; downgrade D→I where chain is missing. Multi-hour audit.

Status: K56 candidate, will file as separate audit doc.

K57 candidate: Bridge Object tier definition (Cal flag b)

Section 5.10 introduced Bridge Object as a new architectural tier (Grace's category, K3 + 49a1 + Q⁵ as Bridge Objects). Cal flag: is Bridge Object genuinely separate from "I-tier observation under an existing L1 Root," or is this tier proliferation?

Audit scope: scrutinize each Bridge Object (K3, 49a1, Q⁵) for whether it ANCHORS independently of its L1 root, or merely COLLECTS L1-source-connections at one mathematical object. Architectural-decision audit.

Status: K57 candidate, will file as separate audit doc.

K58 candidate: Type C strict-protocol spot-audit (Cal flag c)

Random 3-5 N-way density Type C clusters checked under Grace's P1-P7 strict context-counting protocol. Verify density claims hold under strict counting.

Audit scope: select 3-5 random Type C clusters from the catalog (e.g., from Grace's Toy 3074 8-item ≥ 4 -context list, or earlier Type C density-rule findings), apply P1-P7 strict counting protocol, report whether density claims survive.

Status: K58 candidate, will file as separate audit doc. Lower priority than K56/K57.

Mode 7 methodology extension — ENDORSED

Cal proposed adding Mode 7 to Coincidence_Filter_Risk methodology: "classical-integer-set source claim without mechanism chain" as standing audit trigger.

This forward-prevents Heegner-style walk-backs at drafting stage. New L1-source proposals must exhibit a classical-math mechanism chain (W3 negation) before tier label "Root #N" or "established L1 classical source" can be applied.

I endorse Mode 7 as standing methodology extension. Cal+Grace already co-authored Mode 6 corollary; Mode 7 builds on that pattern. Grace files Mode 7 addition to notes/BST_Methodology_Coincidence_Filter_Risk.md per same process as Mode 6.

Per Casey's standard

- Simple: four criteria (W1-W4), three-tier output list (A/B/C), one new methodology mode (Mode 7).

- Works: criteria fire correctly on the Heegner case (the walk-back was needed); pass cleanly on the established roots (no false positives identified).
- Hard to break: would require a classical-integer-set claim that passes W1-W4 negative test but should still walk back, OR a claim that fires W1-W4 positively but doesn't need walk-back. None identified at this scoping.
- Counter-example: the 9 ESTABLISHED roots all pass cleanly (Tier A) — confirmation, not counter-example.

K55 verdict: STRUCTURAL AUDIT RULING

Four criteria ratified. Tier-A clean (9 items, no action). Tier-B verify (4 items, ~1-2h spot-check). Tier-C walk-back (3 candidates → K56, K57, K58 follow-up audits). Mode 7 methodology extension endorsed.

No L1-source promotions are walked back by K55 itself. K47 Heegner-Stark stays D-tier (mechanism via Deuring CM + 49a1 holds). The 9 ESTABLISHED L1 sources stay established. K55 ratifies the criteria for distinguishing them from would-be claims that haven't earned the tier.

The Sunday walk-back of "9 Heegner discriminants as L1 source" stays walked-back. K55 makes that walk-back's criteria explicit.

Action items

1. Grace catalog Mode 7 addition to notes/BST_Methodology_Coincidence_Filter_Risk.md (parallel process to Mode 6). Cal+Grace+Keeper co-authored.
2. Tier-B spot-check (1-2h, distributed): Grace or Cal can take 1-4 items as time permits. Output: 4 ☐ ratifications or specific concerns to refer to K-audit.
3. K56 audit (Cathedral component-level Δr table): Keeper files separately today.
4. K57 audit (Bridge Object tier definition): Keeper files separately today.
5. K58 audit (Type C strict-protocol spot-check): Keeper files at lower priority, scope-limited.
6. Standing rule: future L1-source proposals from any team member must explicitly pass W3 negation (mechanism chain exhibited) before tier label "Root #N" or "ESTABLISHED L1" applied. Casey-approved if he ratifies via standing-directive.

K55 status

Structural audit ruling filed. Four-criteria framework established as standing audit tool. Tier-A 9 ratified, Tier-B 4 flagged for spot-check, Tier-C 3 referred to K56-K58. Mode 7 methodology extension endorsed. Honors Grace's Sunday self-withdrawal precedent and Cal's Tuesday morning scoping work.

— Keeper, 2026-05-19 Tuesday 10:30 EDT